Chapter 6

Water Procurement

Water is one of your most urgent needs in a survival situation. You can't live long without it, especially in hot areas where you lose water rapidly through perspiration. Even in cold areas, you need a minimum of 2 liters of water each day to maintain efficiency.

More than three-fourths of your body is composed of fluids. Your body loses fluid because of heat, cold, stress, and exertion. To function effectively, you must replace the fluid your body loses. So, one of your first goals is to obtain an adequate supply of water.

WATER SOURCES

6-1. Almost any environment has water present to some degree. Figure 6-1, pages 6-2 and 6-3, lists possible sources of water in various environments. It also provides information on how to make the water potable.

NOTE: If you do not have a canteen, cup, can, or other type of container, improvise one from plastic or water-resistant cloth. Shape the plastic or cloth into a bowl by pleating it. Use pins or other suitable items—even your hands—to hold the pleats.

Environment	Sources of Water	Means of Obtaining and/or Making Potable	Remarks
Frigid areas	Snow and ice	Melt and purify.	**Do not** eat without melting! Eating snow or ice can reduce body temperature and lead to more dehydration. Snow or ice are no purer than the water from which they come. Sea ice that is gray in color or opaque is salty. Do not use it without desalting it. Sea ice that is crystalline with a bluish cast has little salt in it.
At sea	Sea	Use desalinator.	**Do not** drink seawater without desalting.
	Rain	Catch rain in tarps or in other water-holding containers.	If tarp or water-holding material is coated with salt, wash it in the sea before using (very little salt will remain on it).
	Sea ice		See previous remarks for frigid areas.
Beach	Ground	Dig hole deep enough to allow water to seep in; obtain rocks, build fire, and heat rocks; drop hot rocks in water; hold cloth over hole to absorb steam; wring water from cloth.	Alternate method if a container or bark pot is available: Fill container or pot with seawater; build fire and boil water to produce steam; hold cloth over container to absorb steam; wring water from cloth.
	Fresh	Dig behind first group of sand dunes. This will allow the collection of fresh water.	
Desert	Ground • In valleys and low areas • At foot of concave banks of dry rivers • At foot of cliffs or rock outcrops • At first depression behind first sand dune of dry lakes • Wherever you find damp surface sand • Wherever you find green vegetation	Dig holes deep enough to allow water to seep in.	In a sand dune belt, any available water will be found beneath the original valley floor at the edge of dunes.

Figure 6-1. Water Sources in Different Environments

Environment	Sources of Water	Means of Obtaining and/or Making Potable	Remarks
Desert (cont)	Cacti	Cut off the top of a barrel cactus and mash or squeeze the pulp. **CAUTION: Do not eat pulp. Place pulp in mouth, suck out juice, and discard pulp.**	Without a machete, cutting into a cactus is difficult and takes time since you must get past the long, strong spines and cut through the tough rind.
	Depressions or holes in rocks		Periodic rainfall may collect in pools, seep into fissures, or collect in holes in rocks.
	Fissures in rock	Insert flexible tubing and siphon water. If fissure is large enough, you can lower a container into it.	
	Porous rock	Insert flexible tubing and siphon water.	
	Condensation on metal	Use cloth to absorb water, then wring water from cloth.	Extreme temperature variations between night and day may cause condensation on metal surfaces. Following are signs to watch for in the desert to help you find water: • All trails lead to water. You should follow in the direction in which the trails converge. Signs of camps, campfire ashes, animal droppings, and trampled terrain may mark trails. • Flocks of birds will circle over water holes. Some birds fly to water holes at dawn and sunset. Their flight at these times is usually fast and close to the ground. Bird tracks or chirping sounds in the evening or early morning sometimes indicate that water is nearby.

Figure 6-1. Water Sources in Different Environments (Continued)

6-2. If you do not have a reliable source to replenish your water supply, stay alert for ways in which your environment can help you.

NOTE: DO NOT substitute the fluids listed in Figure 6-2 for water.

Fluid	Remarks
Alcoholic beverages	Dehydrate the body and cloud judgment.
Urine	Contains harmful body wastes. Is about 2 percent salt.
Blood	Is salty and considered a food; therefore, requires additional body fluids to digest. May transmit disease.
Seawater	Is about 4 percent salt. It takes about 2 liters of body fluids to rid the body of waste from 1 liter of seawater. Therefore, by drinking seawater you deplete your body's water supply, which can cause death.

Figure 6-2. The Effects of Substitute Fluids

6-3. Heavy dew can provide water. Tie rags or tufts of fine grass around your ankles and walk through dew-covered grass before sunrise. As the rags or grass tufts absorb the dew, wring the water into a container. Repeat the process until you have a supply of water or until the dew is gone. Australian natives sometimes mop up as much as 1 liter an hour this way.

6-4. Bees or ants going into a hole in a tree may point to a water-filled hole. Siphon the water with plastic tubing or scoop it up with an improvised dipper. You can also stuff cloth in the hole to absorb the water and then wring it from the cloth.

6-5. Water sometimes gathers in tree crotches or rock crevices. Use the above procedures to get the water. In arid areas, bird droppings around a crack in the rocks may indicate water in or near the crack.

6-6. Green bamboo thickets are an excellent source of fresh water. Water from green bamboo is clear and odorless. To get the water, bend a green bamboo stalk, tie it down, and cut off the top (Figure 6-3, page 6-5). The water will drip freely during the night. Old, cracked bamboo may also contain water.



Figure 6-3. Water From Green Bamboo

CAUTION

Purify the water before drinking it.

6-7. Wherever you find banana trees, plantain trees, or sugarcane, you can get water. Cut down the tree, leaving about a 30-centimeter (12-inch) stump, and scoop out the center of the stump so that the hollow is bowl-shaped. Water from the roots will immediately start to fill the hollow. The first three fillings of water will be bitter, but succeeding fillings will be palatable. The stump (Figure 6-4, page 6-6) will supply water for up to 4 days. Be sure to cover it to keep out insects.

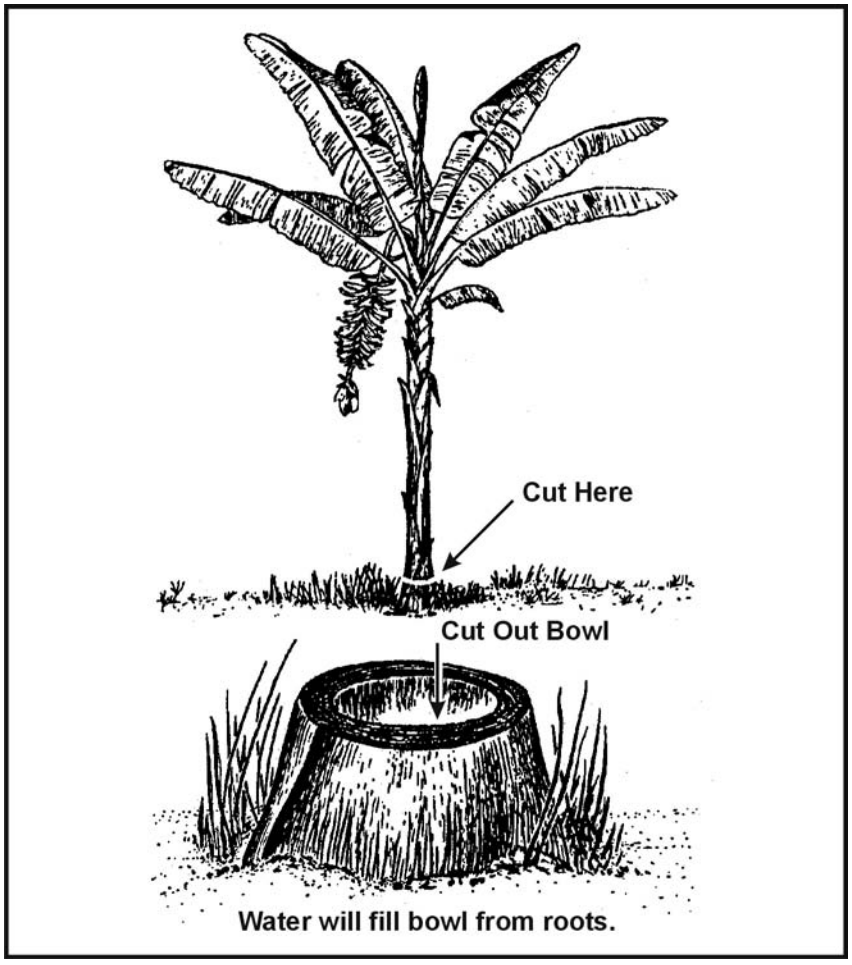


Figure 6-4. Water From Plantain or Banana Tree Stump

6-8. Some tropical vines can give you water. Cut a notch in the vine as high as you can reach, then cut the vine off close to the ground. Catch the dropping liquid in a container or in your mouth (Figure 6-5, page 6-7).

CAUTION
Ensure that the vine is not poisonous.

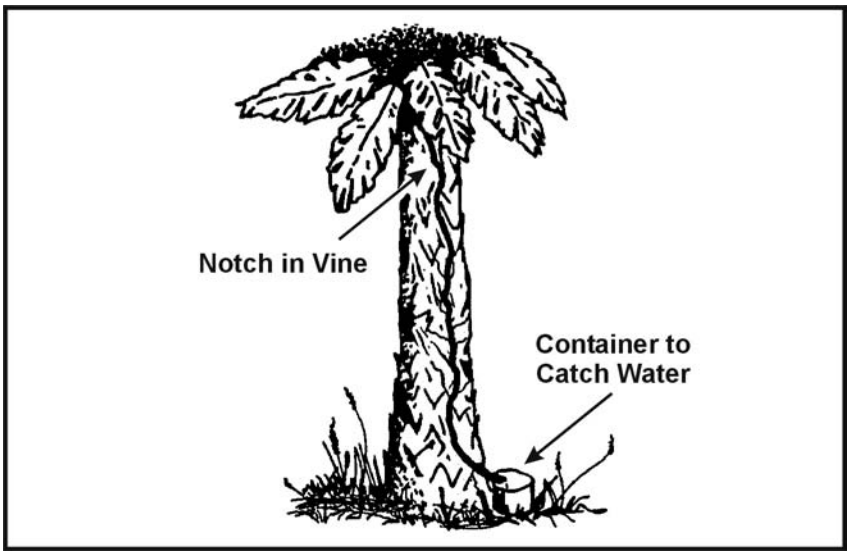


Figure 6-5. Water From a Vine

6-9. The milk from young, green (unripe) coconuts is a good thirst quencher. However, the milk from mature, brown, coconuts contains an oil that acts as a laxative. Drink in moderation only.

CAUTION

Do not drink the liquid if it is sticky, milky, or bitter tasting.

6-10. In the American tropics you may find large trees whose branches support air plants. These air plants may hold a considerable amount of rainwater in their overlapping, thickly growing leaves. Strain the water through a cloth to remove insects and debris.

6-11. You can get water from plants with moist pulpy centers. Cut off a section of the plant and squeeze or smash the pulp so that the moisture runs out. Catch the liquid in a container.

6-12. Plant roots may provide water. Dig or pry the roots out of the ground, cut them into short pieces, and smash the pulp so that the moisture runs out. Catch the liquid in a container.

6-13. Fleshy leaves, stems, or stalks, such as bamboo, contain water. Cut or notch the stalks at the base of a joint to drain out the liquid.

6-14. The following trees can also provide water:

- *Palms.* The buri, coconut, sugar, rattan, and nips contain liquid. Bruise a lower frond and pull it down so the tree will "bleed" at the injury.
- *Traveler's tree.* Found in Madagascar, this tree has a cuplike sheath at the base of its leaves in which water collects.
- *Umbrella tree.* The leaf bases and roots of this tree of western tropical Africa can provide water.
- *Baobab tree.* This tree of the sandy plains of northern Australia and Africa collects water in its bottlelike trunk during the wet season. Frequently, you can find clear, fresh water in these trees after weeks of dry weather.

CAUTION

Do not keep the sap from plants longer than 24 hours. It begins fermenting, becoming dangerous as a water source.

STILL CONSTRUCTION

6-15. You can use stills in various areas of the world. They draw moisture from the ground and from plant material. You need certain materials to build a still, and you need time to let it collect the water. It takes about 24 hours to get 0.5 to 1 liter of water.

ABOVEGROUND STILLS

6-16. You can construct two types of aboveground stills. To make the **vegetation bag still**, you need a sunny slope on which to place the still, a clear plastic bag, green leafy vegetation, and a small rock (Figure 6-6, page 6-9).

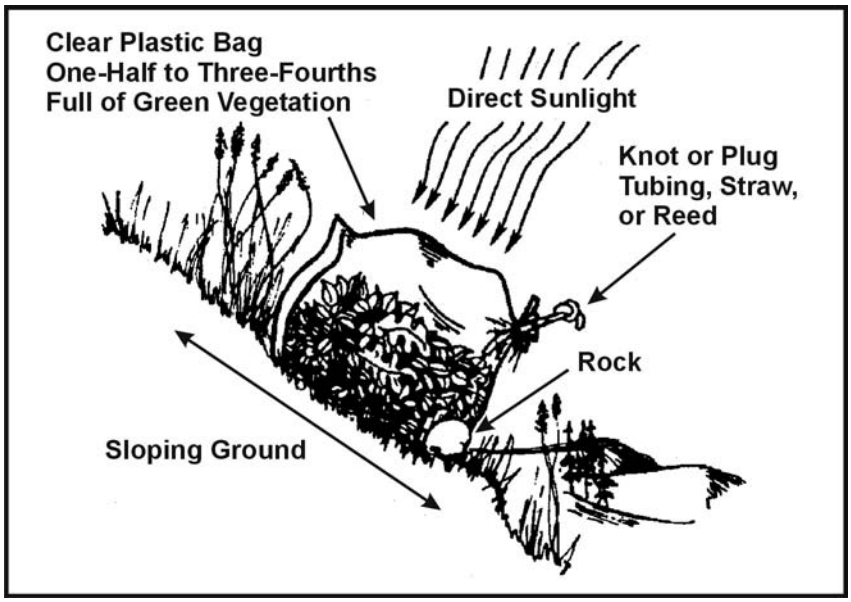


Figure 6-6. Vegetation Bag Still

6-17. To make the still, you should—

- Fill the bag with air by turning the opening into the breeze or by "scooping" air into the bag.
- Fill the plastic bag one-half to three-fourths full of green leafy vegetation. Be sure to remove all hard sticks or sharp spines that might puncture the bag.

CAUTION

Do not use poisonous vegetation. It will provide poisonous liquid.

- Place a small rock or similar item in the bag.
- Close the bag and tie the mouth securely as close to the end of the bag as possible to keep the maximum amount of air space. If you have a piece of tubing, a small straw, or a hollow reed, insert one end in the mouth of the bag before you tie it securely. Then tie off or plug the tubing so that

air will not escape. This tubing will allow you to drain out condensed water without untying the bag.

- Place the bag, mouth downhill, on a slope in full sunlight. Position the mouth of the bag slightly higher than the low point in the bag.
- Settle the bag in place so that the rock works itself into the low point in the bag.

6-18. To get the condensed water from the still, loosen the tie around the bag's mouth and tip the bag so that the water collected around the rock will drain out. Then retie the mouth securely and reposition the still to allow further condensation.

6-19. Change the vegetation in the bag after extracting most of the water from it. This will ensure maximum output of water.

6-20. Making a **transpiration bag still** is similar to the vegetation bag, only easier. Simply tie the plastic bag over a leafy tree limb with a tube inserted, and tie the mouth of the bag off tightly around the branch to form an airtight seal. Tie the end of the limb so that it hangs below the level of the mouth of the bag. The water will collect there (Figure 6-7).

6-21. The same limb may be used for 3 to 5 days without causing long-term harm to the limb. It will heal itself within a few hours of removing the bag.



Figure 6-7. Water Transpiration Bag

BELOWGROUND STILL

6-22. To make a belowground still, you need a digging tool, a container, a clear plastic sheet, a drinking tube, and a rock (Figure 6-8).

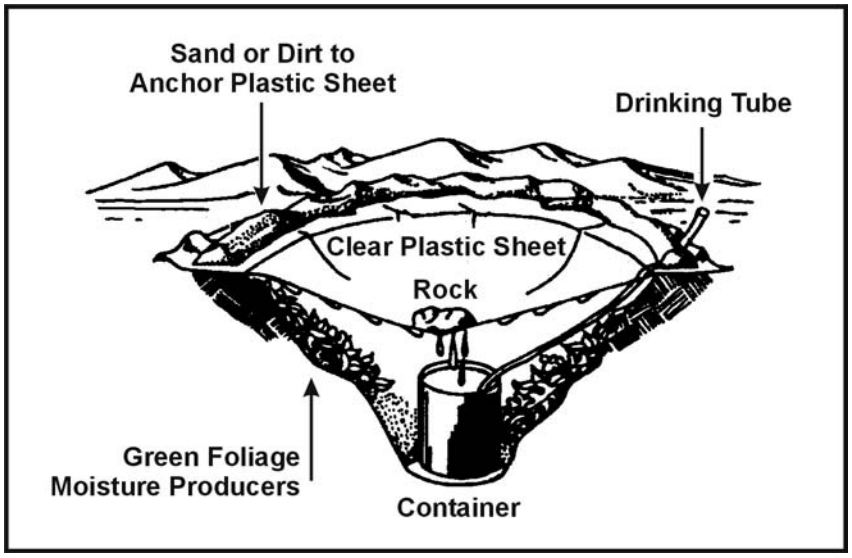


Figure 6-8. Belowground Still

6-23. Select a site where you believe the soil will contain moisture (such as a dry streambed or a low spot where rainwater has collected). The soil at this site should be easy to dig, and sunlight must hit the site most of the day.

6-24. To construct the still, you should—

- Dig a bowl-shaped hole about 1 meter (3 feet) across and 60 centimeters (24 inches) deep.
- Dig a sump in the center of the hole. The sump's depth and perimeter will depend on the size of the container that you have to place in it. The bottom of the sump should allow the container to stand upright.
- Anchor the tubing to the container's bottom by forming a loose overhand knot in the tubing.
- Place the container upright in the sump.

- Extend the unanchored end of the tubing up, over, and beyond the lip of the hole.
- Place the plastic sheet over the hole, covering its edges with soil to hold it in place.
- Place a rock in the center of the plastic sheet.
- Lower the plastic sheet into the hole until it is about 40 centimeters (16 inches) below ground level. It now forms an inverted cone with the rock at its apex. Make sure that the cone's apex is directly over your container. Also make sure the plastic cone does not touch the sides of the hole because the earth will absorb the condensed water.
- Put more soil on the edges of the plastic to hold it securely in place and to prevent the loss of moisture.
- Plug the tube when not in use to keep the moisture from evaporating and to keep insects out.

6-25. You can drink water without disturbing the still by using the tube as a straw. By opening the still, you release the moist, warm air that has accumulated.

6-26. You may want to use plants in the hole as a moisture source. If so, dig out additional soil from the sides of the hole to form a slope on which to place the plants. Then proceed as above.

6-27. If polluted water is your only moisture source, dig a small trough outside the hole about 25 centimeters (10 inches) from the still's lip (Figure 6-9, page 6-13). Dig the trough about 25 centimeters (10 inches) deep and 8 centimeters (3 inches) wide. Pour the polluted water in the trough. Be sure you do not spill any polluted water around the rim of the hole where the plastic sheet touches the soil. The trough holds the polluted water and the soil filters it as the still draws it. The water then condenses on the plastic and drains into the container. This process works extremely well when your only water source is salt water.

6-28. You will need at least three stills to meet your individual daily water intake needs. In comparison to the belowground still and the water transpiration bag still, the vegetation bag produces the best yield of water.

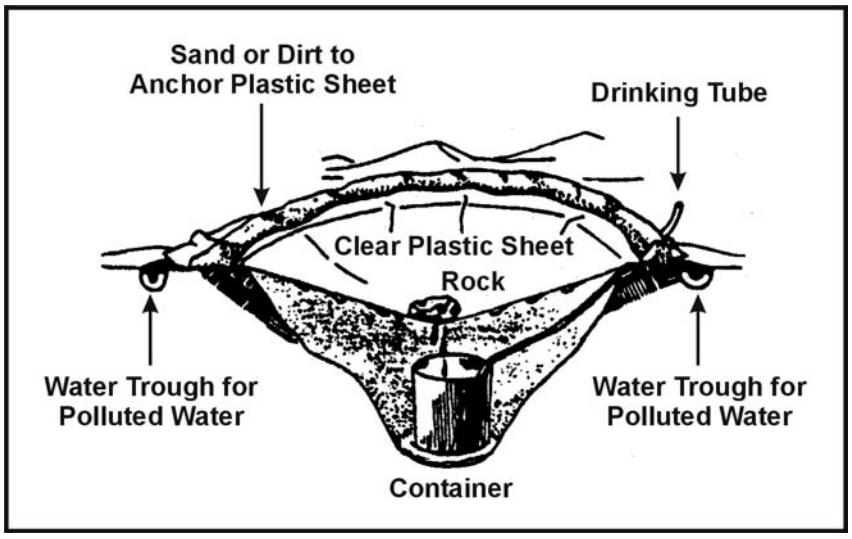


Figure 6-9. Belowground Still to Get Potable Water From Polluted Water

WATER PURIFICATION

6-29. Rainwater collected in clean containers or in plants is usually safe for drinking. However, purify water from lakes, ponds, swamps, springs, or streams, especially the water near human settlements or in the tropics.

6-30. When possible, purify all water you get from vegetation or from the ground by boiling or using iodine or chlorine. After purifying a canteen of water, you must partially unscrew the cap and turn the canteen upside down to rinse unpurified water from the threads of the canteen where your mouth touches.

6-31. Purify water by the following methods:

- Use water purification tablets. (Follow the directions provided.)
- Place 5 drops of 2 percent tincture of iodine in a canteen full of clear water. If the canteen is full of cloudy or cold water, use 10 drops. (Let the canteen of water stand for 30 minutes before drinking.)
- Use 2 drops of 10 percent (military strength) povidone-iodine or 1 percent titrated povidone-iodine. The civilian

equivalent is usually 2 percent strength, so 10 drops will be needed. Let stand for 30 minutes. If the water is cold and clear, wait 60 minutes. If it's very cold or cloudy, add 4 drops and wait 60 minutes.

- Place 2 drops of chlorine bleach (5.25 percent sodium hypochlorite) in a canteen of water. Let stand 30 minutes. If the water is cold or cloudy, wait 60 minutes. Remember that not all bleach is the same around the world; check the available level of sodium hypochlorite.
- Use potassium permanganate, commonly marketed as Condy's Crystals, for a number of applications, including emergency disinfection of water. The crystals are of a nonuniform size, so you must judge the actual dosage by the color of the water after adding the crystals. Add three small crystals to 1 liter (1 quart) of water. If the water turns a bright pink after waiting 30 minutes, the water is considered purified. If the water turns a dark pink, there is too much potassium permanganate to drink safely. Either add more water to dilute the mixture or save it for use as an antiseptic solution. If the water becomes a full red, like the color of cranberry juice, the solution may be used as an antifungal solution.
- Boil your drinking water. This is the safest method of purifying your drinking water. By achieving a rolling boil, you can ensure that you are destroying all living waterborne pathogens.

6-32. By drinking nonpotable water you may contract diseases or swallow organisms that can harm you and may easily lead to potentially fatal waterborne illnesses.

6-33. Two of the most prevalent pathogens found in most water sources throughout the world are—

- Giardia, which causes Giardiasis (beaver fever). It is characterized by an explosive, watery diarrhea accompanied by severe cramps lasting 7 to 14 days.
- Cryptosporidium, which causes Cryptosporidiosis. It is much like Giardiasis, only more severe and prolonged, and there is no known cure but time. Diarrhea may be mild and can last from 3 days to 2 weeks.

NOTE: The only effective means of neutralizing Cryptosporidium is by boiling or by using a commercial microfilter or reverse-osmosis filtration system. Chemical disinfectants such as iodine tablets or bleach have not shown to be 100 percent effective in eliminating Cryptosporidium.

6-34. Examples of other diseases or organisms are—

- *Dysentery.* You may experience severe, prolonged diarrhea with bloody stools, fever, and weakness.
- *Cholera and typhoid.* You may be susceptible to these diseases regardless of inoculations. Cholera can cause profuse, watery diarrhea, vomiting, and leg cramps. Typhoid symptoms include fever, headache, loss of appetite, constipation, and bleeding in the bowel.
- *Hepatitis A.* Symptoms include diarrhea, abdominal pain, jaundice, and dark urine. This infection can spread through close person-to-person contact or ingestion of contaminated water or food.
- *Flukes.* Stagnant, polluted water—especially in tropical areas—often contains blood flukes. If you swallow flukes, they will bore into the bloodstream, live as parasites, and cause disease.
- *Leeches.* If you swallow a leech, it can hook onto the throat passage or inside the nose. It will suck blood, create a wound, and move to another area. Each bleeding wound may become infected.

WATER FILTRATION DEVICES

6-35. If the water you find is also muddy, stagnant, and foul-smelling, you can clear the water—

- By placing it in a container and letting it stand for 12 hours.
- By pouring it through a filtering system.

NOTE: These procedures only clear the water and make it more palatable. You will have to purify it.

6-36. To make a filtering system, place several centimeters or layers of filtering material such as sand, crushed rock, charcoal,

or cloth in bamboo, a hollow log, or an article of clothing (Figure 6-10).

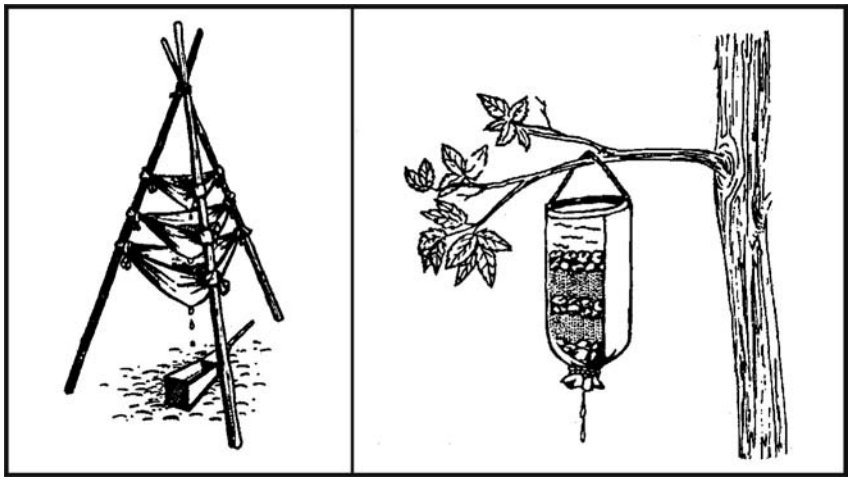


Figure 6-10. Water Filtering Systems

6-37. Remove the odor from water by adding charcoal from your fire. Charcoal is also helpful in absorbing some agricultural and industrial chemicals. Let the water stand for 45 minutes before drinking it.

Chapter 7

Firecraft

In many survival situations, the ability to start a fire can make the difference between living and dying. Fire can fulfill many needs. It can provide warmth and comfort. It not only cooks and preserves food, it also provides warmth in the form of heated food that saves calories our body normally uses to produce body heat. You can use fire to purify water, sterilize bandages, signal for rescue, and provide protection from animals. It can be a psychological boost by providing peace of mind and companionship. You can also use fire to produce tools and weapons.

Fire can cause problems, as well. The enemy can detect the smoke and light it produces. It can cause forest fires or destroy essential equipment. Fire can also cause burns and carbon monoxide poisoning when used in shelters.

Weigh your need for fire against your need to avoid enemy detection.

BASIC FIRE PRINCIPLES

7-1. To build a fire, it helps to understand the basic principles of a fire. Fuel (in a nongaseous state) does not burn directly. When you apply heat to a fuel, it produces a gas. This gas, combined with oxygen in the air, burns.

7-2. Understanding the concept of the fire triangle is very important in correctly constructing and maintaining a fire. The three sides of the triangle represent **air, heat,** and **fuel**. If you remove any of these, the fire will go out. The correct ratio of these components is very important for a fire to burn at its greatest capability. The only way to learn this ratio is to practice.

SITE SELECTION AND PREPARATION

7-3. You will have to decide what site and arrangement to use. Before building a fire consider—

- The area (terrain and climate) in which you are operating.
- The materials and tools available.
- Time; how much time do you have?
- Need; why do you need a fire?
- Security; how close is the enemy?

7-4. Look for a dry spot that—

- Is protected from the wind.
- Is suitably placed in relation to your shelter (if any).
- Will concentrate the heat in the direction you desire.
- Has a supply of wood or other fuel available. (Figure 7-4, pages 7-5 and 7-6, lists types of material you can use.)

7-5. If you are in a wooded or brush-covered area, clear the brush and scrape the surface soil from the spot you have selected. Clear a circle at least 1 meter (3 feet) in diameter so there is little chance of the fire spreading.

7-6. If time allows, construct a fire wall using logs or rocks. This wall will help to reflect or direct the heat where you want it (Figure 7-1, page 7-3). It will also reduce flying sparks and cut down on the amount of wind blowing into the fire. However, you will need enough wind to keep the fire burning.

CAUTION

Do not use wet or porous rocks as they may explode when heated.

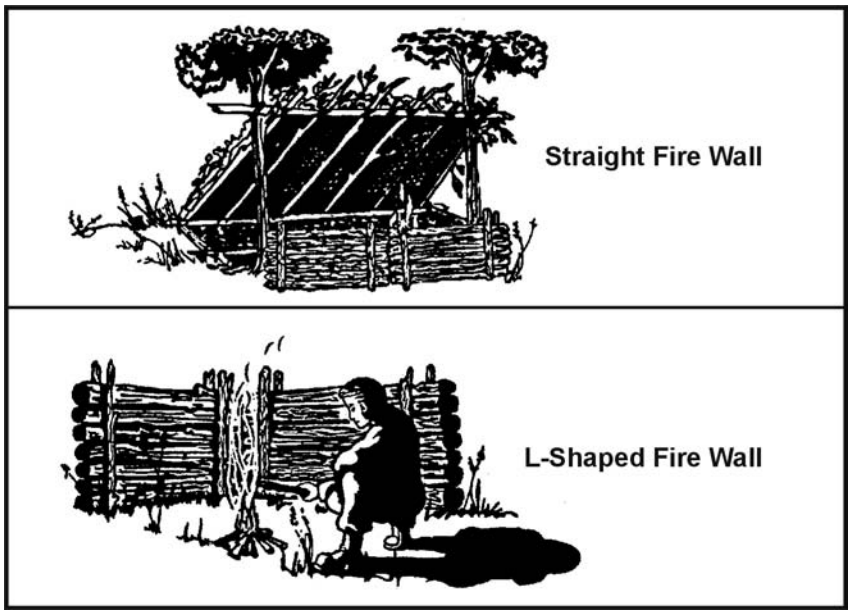


Figure 7-1. Types of Fire Walls

7-7. In some situations, you may find that an underground fireplace will best meet your needs. It conceals the fire and serves well for cooking food. To make an underground fireplace or Dakota fire hole (Figure 7-2, page 7-4)—

- Dig a hole in the ground.
- On the upwind side of this hole, poke or dig a large connecting hole for ventilation.
- Build your fire in the hole as illustrated.

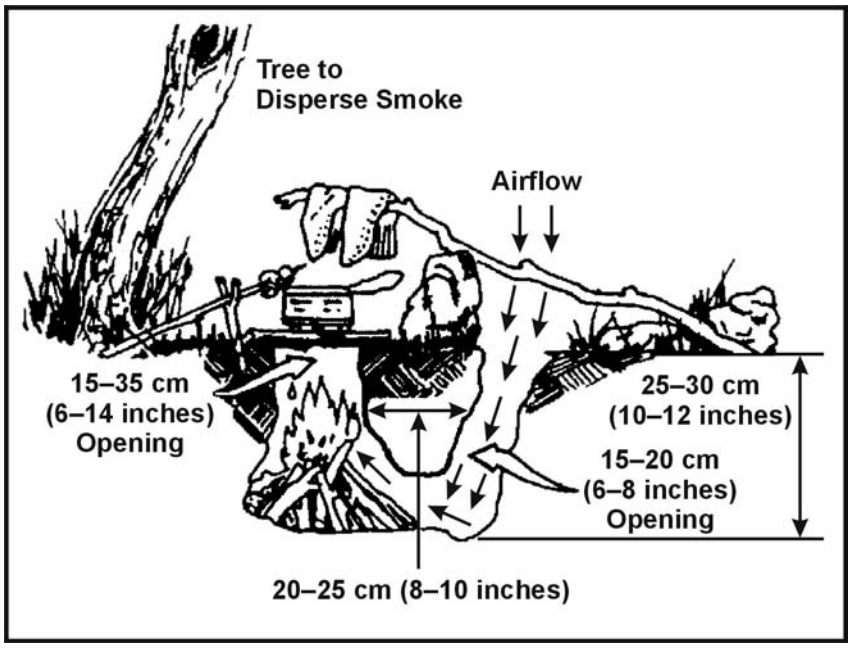


Figure 7-2. Dakota Fire Hole

7-8. If you are in a snow-covered area, use green logs to make a dry base for your fire (Figure 7-3). Trees with wrist-sized trunks are easily broken in extreme cold. Cut or break several green logs and lay them side by side on top of the snow. Add one or two more layers. Lay the top layer of logs opposite those below it.

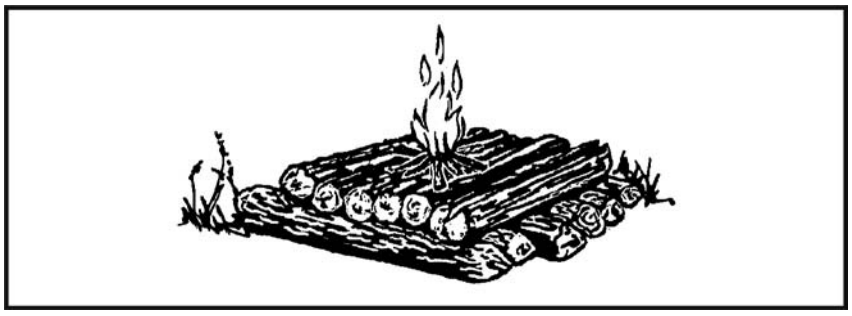


Figure 7-3. Base for Fire in Snow-covered Area

FIRE MATERIAL SELECTION

7-9. You need three types of materials (Figure 7-4, pages 7-5 and 7-6) to build a fire.

7-10. Tinder is dry material that ignites with little heat—a spark starts a fire. The tinder must be absolutely dry to be sure just a spark will ignite it. If you have a device that generates only sparks, charred cloth will be almost essential. It holds a spark for long periods, allowing you to put tinder on the hot area to generate a small flame. You can make charred cloth by heating cotton cloth until it turns black, but does not burn. Once it is black, you must keep it in an airtight container to keep it dry. Prepare this cloth well in advance of any survival situation. Add it to your individual survival kit. Other impromptu items could be alcohol pads or petroleum jelly gauze.

7-11. Kindling is readily combustible material that you add to the burning tinder. Again, this material should be absolutely dry to ensure rapid burning. Kindling increases the fire's temperature so that it will ignite less combustible material.

7-12. Fuel is less combustible material that burns slowly and steadily once ignited.

Tinder	Kindling	Fuel
• Birch bark. • Shredded inner bark from cedar, chestnut, red elm trees. • Fine wood shavings. • Dead grass, ferns, moss, fungi. • Straw. • Sawdust. • Very fine pitchwood scrapings. • Dead evergreen needles.	• Small twigs. • Small strips of wood. • Lighter knot from pine tree stumps with a heavy concentration of resin. • Heavy cardboard. • Pieces of wood removed from the inside of larger pieces. • Wood that has been doused with highly flammable materials, such as gasoline, oil, or wax.	• Dry, standing wood and dry, dead branches. • Dry inside (heart) of fallen tree trunks and large branches. • Green wood that is finely split. • Dry grasses twisted into bunches.

Figure 7-4. Materials for Building Fires

Tinder	Kindling	Fuel
• Punk (the completely rotted portions of dead logs or trees). • Evergreen tree knots. • Bird down (fine feathers). • Down seed heads (milkweed, dry cattails, bulrush, or thistle). • Fine, dried vegetable fibers. • Spongy threads of dead puffball. • Dead palm leaves. • Skinlike membrane lining bamboo. • Lint from pockets and seams. • Charred cloth. • Waxed paper. • Other bamboo shavings. • Gunpowder. • Cotton.		• Peat dry enough to burn (this may be found at the top of undercut banks). • Dried animal dung. • Animal fats. • Coal, oil shale, or oil lying on the surface.

Figure 7-4. Materials for Building Fires (Continued)

HOW TO BUILD A FIRE

7-13. There are several methods for laying a fire and each one has advantages. The situation you are in will determine which of the following fires to use.

TEPEE

7-14. To make a tepee fire (Figure 7-5, page 7-7), arrange the tinder and a few sticks of kindling in the shape of a tepee or cone. Light the center. As the tepee burns, the outside logs will fall inward, feeding the fire. This type of fire burns well even with wet wood.

LEAN-TO

7-15. To lay a lean-to fire (Figure 7-5), push a green stick into the ground at a 30-degree angle. Point the end of the stick in the direction of the wind. Place some tinder deep under this lean-to stick. Lean pieces of kindling against the lean-to stick. Light the tinder. As the kindling catches fire from the tinder, add more kindling.

CROSS-DITCH

7-16. To use the cross-ditch method (Figure 7-5), scratch a cross about 30 centimeters (12 inches) in size in the ground. Dig the cross 7.5 centimeters (about 3 inches) deep. Put a large wad of tinder in the middle of the cross. Build a kindling pyramid above the tinder. The shallow ditch allows air to sweep under the tinder to provide a draft.

PYRAMID

7-17. To lay the pyramid fire (Figure 7-5), place two small logs or branches parallel on the ground. Place a solid layer of small logs across the parallel logs. Add three or four more layers of logs, each layer smaller than and at a right angle to the layer below it. Make a starter fire on top of the pyramid. As the starter fire burns, it will ignite the logs below it. This gives you a fire that burns downward, requiring no attention during the night.

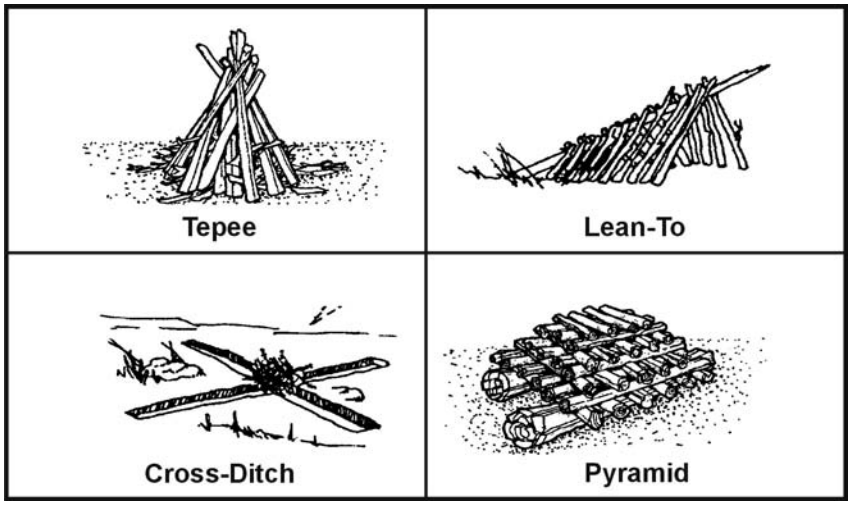


Figure 7-5. Methods for Laying Fires

7-18. There are several other ways to lay a fire that are quite effective. Your situation and the material available in the area may make another method more suitable.

HOW TO LIGHT A FIRE

7-19. Always light your fire from the upwind side. Make sure you lay the tinder, kindling, and fuel so that your fire will burn as long as you need it. Igniters provide the initial heat required to start the tinder burning. They fall into two categories: modern methods and primitive methods.

MODERN METHODS

7-20. Modern igniters use modern devices. These are items that we normally think of to start a fire.

Matches

7-21. Make sure these matches are waterproof. Also, store them in a waterproof container along with a dependable striker pad.

Convex Lens

7-22. Use this method (Figure 7-6) only on bright, sunny days. The lens can come from binoculars, a camera, telescopic sights, or magnifying glasses. Angle the lens to concentrate the sun's rays on the tinder. Hold the lens over the same spot until the tinder begins to smolder. Gently blow or fan the tinder into a flame and apply it to the fire lay.

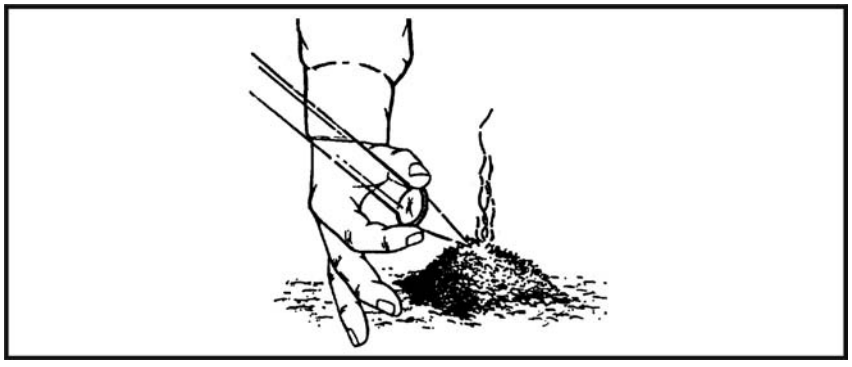


Figure 7-6. Lens Method

Metal Match

7-23. Place a flat, dry leaf under your tinder with a portion exposed. Place the tip of the metal match on the dry leaf, holding the metal match in one hand and a knife in the other. Scrape your knife against the metal match to produce sparks. The sparks will hit the tinder. When the tinder starts to smolder, proceed as above.

Battery

7-24. Use a battery to generate a spark. Use of this method depends on the type of battery available. Attach a wire to each terminal. Touch the ends of the bare wires together next to the tinder so the sparks will ignite it.

Gunpowder

7-25. Often, you will have ammunition with your equipment. If so, carefully extract the bullet from the shell casing by moving the bullet back and forth. Use the gunpowder as tinder. Discard the casing and primers. A spark will ignite the powder.

NOTE: Be extremely careful during this operation as the primers are still sensitive and even a small pile of gunpowder can give surprising results.

PRIMITIVE METHODS

7-26. Primitive igniters are those attributed to our early ancestors. They can be time-consuming, which requires you to be patient and persistent.

Flint and Steel

7-27. The direct spark method is the easiest of the primitive methods to use. The flint and steel method is the most reliable of the direct spark methods. Strike a flint or other hard, sharp-edged rock with a piece of carbon steel (stainless steel will not produce a good spark). This method requires a loose-jointed wrist and practice. When the tinder catches a spark, blow on it. The spark will spread and burst into flames.

Fire-Plow

7-28. The fire-plow (Figure 7-7) is a friction method of ignition. To use this method, cut a straight groove in a softwood base and plow the blunt tip of a hardwood shaft up and down the groove. The plowing action of the shaft pushes out small particles of wood fibers. Then, as you apply more pressure on each stroke, the friction ignites the wood particles.

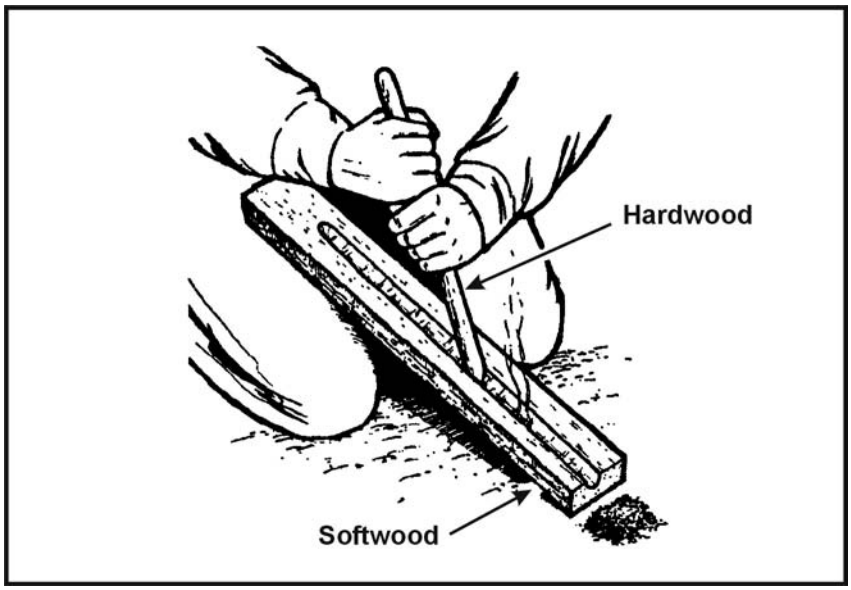


Figure 7-7. Fire-Plow

Bow and Drill

7-29. The technique of starting a fire with a bow and drill (Figure 7-8, page 7-11) is simple, but you must exert much effort and be persistent to produce a fire. You need the following items to use this method:

- *Socket.* The socket is an easily grasped stone or piece of hardwood with a slight depression in one side. Use it to hold the drill in place and to apply downward pressure.
- *Drill.* The drill should be a straight, seasoned hardwood stick about 2 centimeters (3/4 inch) in diameter and 25 centimeters (10 inches) long. The top end is round and the low end blunt (to produce more friction).

- *Fire board.* Although any board may be used, a seasoned softwood board about 2.5 centimeters (1 inch) thick and 10 centimeters (4 inches) wide is preferable. Cut a depression about 2 centimeters (3/4 inch) from the edge on one side of the board. On the underside, make a V-shaped cut from the edge of the board to the depression.
- *Bow.* The bow is a resilient, green stick about 2.5 centimeters (3/4 inch) in diameter with a bowstring. The type of wood is not important. The bowstring can be any type of cordage. Tie the bowstring from one end of the bow to the other, without any slack.

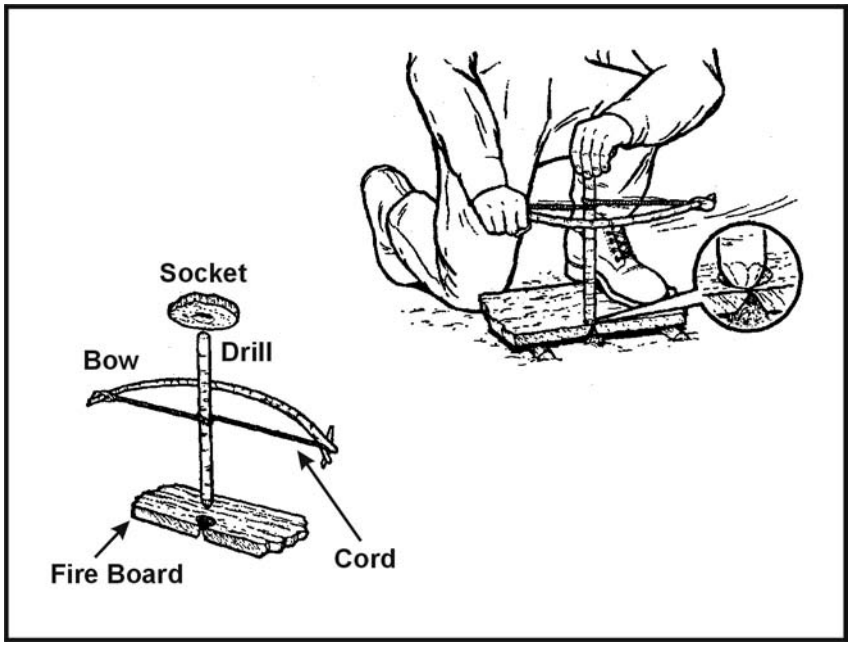


Figure 7-8. Bow and Drill

7-30. To use the bow and drill, first prepare the fire lay. Then place a bundle of tinder under the V-shaped cut in the fire board. Place one foot on the fire board. Loop the bowstring over the drill and place the drill in the precut depression on the fire board. Place the socket, held in one hand, on the top of the drill to hold it in position. Press down on the drill and saw the bow back and forth to twirl the drill (Figure 7-8). Once you have established a

smooth motion, apply more downward pressure and work the bow faster. This action will grind hot black powder into the tinder, causing a spark to catch. Blow on the tinder until it ignites.

7-31. Primitive fire-building methods are exhausting and require practice to ensure success. If your survival situation requires the use of primitive methods, remember the following hints to help you construct and maintain the fire:

- If possible, use nonaromatic seasoned hardwood for fuel.
- Collect kindling and tinder along the trail.
- Add insect repellent to the tinder.
- Keep the firewood dry.
- Dry damp firewood near the fire.
- Bank the fire to keep the coals alive overnight.
- Carry lighted punk, when possible.
- Be sure the fire is out before leaving camp.
- Do not select wood lying on the ground. It may appear to be dry but generally doesn't provide enough friction.